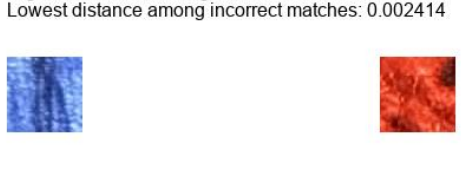
```
Arguments used:
  folder: C:\Users\pensf\Documents\AutomatedLBP_Benchmarking\LBP_Test_Images\Swatches\LBP_Texture_Swatches
  P: 8
  R: 1.0
  method: ror
  metric: chi2
  save_csv: matches.csv
  blur: 0.0
  ltp: False
  visualize: True
  X: 64
  Y: 64
  V: True
  G: False
Correct matches: 45/75 (60.00%)
Highest distance among correct matches: 0.012321
Lowest distance among correct matches: 0.001617
Highest distance among incorrect matches: 0.006256
Lowest distance among incorrect matches: 0.002414
```



```
Index: 0
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.003702
```

```
Index: 1
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.003952
```

```
Index: 2
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.003213
```

```
Index: 3
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.003160
```

```
Index: 4
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.004038
```

```
Index: 5
-- Original --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.002606
```

```
Index: 6
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Chi Squared
Distance: 0.004651
```

```
Index: 7
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Chi Squared
Distance: 0.003167
```

```
Index: 8
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.005877
```

```
Index: 9
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Chi Squared
Distance: 0.005002
```

```
Index: 10
-- Original --
INSTANCE: brown
CATEGORY: brown
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: brown
Distance Metric: Chi Squared
Distance: 0.004651
```

```
Index: 11
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.004993
```

```
Index: 12
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.005207
```

```
Index: 13
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.004993
```

```
Index: 14
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.002429
```

```
Index: 15
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.002414
```

```
Index: 16
-- Original --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.002760
```

```
Index: 17
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.003975
```

```
Index: 18
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Chi Squared
Distance: 0.003006
```

```
Index: 19
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Chi Squared
Distance: 0.003006
```

```
Index: 20
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Chi Squared
Distance: 0.004515
```

```
Index: 21
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.006256
```

```
Index: 22
-- Original --
INSTANCE: gray
CATEGORY: gray
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Chi Squared
Distance: 0.006713
```

```
Index: 23
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.007202
```

```
Index: 24
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.005358
```

```
Index: 25
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.006658
```

```
Index: 26
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: green
-- Matched --
INSTANCE: green
CATEGORY: green
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: green
Distance Metric: Chi Squared
Distance: 0.005358
```

```
Index: 27
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.004239
```

```
Index: 28
-- Original --
INSTANCE: green
CATEGORY: green
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
Distance Metric: Chi Squared
Distance: 0.004513
```

```
Index: 29
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.003752
```

```
Index: 30
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.003877
```

```
Index: 31
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.003752
```

```
Index: 32
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.004666
```

```
Index: 33
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.003657
```

```
Index: 34
-- Original --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.002721
```

```
Index: 35
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003275
```

```
Index: 36
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.004038
```

```
Index: 37
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.002601
```

```
Index: 38
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.003553
```

```
Index: 39
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.003865
```

```
Index: 40
-- Original --
INSTANCE: pink
CATEGORY: pink
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: pink
Distance Metric: Chi Squared
Distance: 0.004493
```

```
Index: 41
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.001617
```

```
Index: 42
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.002429
```

```
Index: 43
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.003780
```

```
Index: 44
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.002414
```

```
Index: 45
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.001617
```

```
Index: 46
-- Original --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
-- Matched --
INSTANCE: lightBlue
CATEGORY: lightBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.002721
```

```
Index: 47
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.001974
```

```
Index: 48
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: brown
-- Matched --
INSTANCE: brown
CATEGORY: brown
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.003167
```

```
Index: 49
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: red
Distance Metric: Chi Squared
Distance: 0.001974
```

```
Index: 50
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
-- Matched --
INSTANCE: purple
CATEGORY: purple
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: lightBlue
Distance Metric: Chi Squared
Distance: 0.002787
```

```
Index: 51
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: darkGreen
-- Matched --
INSTANCE: darkGreen
CATEGORY: darkGreen
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.004141
```

```
Index: 52
-- Original --
INSTANCE: red
CATEGORY: red
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: red
-- Matched --
INSTANCE: red
CATEGORY: red
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: purple
Distance Metric: Chi Squared
Distance: 0.003124
```

```
Index: 53
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.004166
```

```
Index: 54
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.004320
```

```
Index: 55
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Chi Squared
Distance: 0.004429
```

```
Index: 56
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
-- Matched --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelBlue
Distance Metric: Chi Squared
Distance: 0.004166
```

```
Index: 57
-- Original --
INSTANCE: sparkelBlue
CATEGORY: sparkelBlue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: gray
-- Matched --
INSTANCE: gray
CATEGORY: gray
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: gray
Distance Metric: Chi Squared
Distance: 0.004386
```

```
Index: 58
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.003970
```

```
Index: 59
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.005275
```

```
Index: 60
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.003970
```

```
Index: 61
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.005721
```

```
Index: 62
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.005422
```

```
Index: 63
-- Original --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
-- Matched --
INSTANCE: sparkelRed
CATEGORY: sparkelRed
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkelRed
Distance Metric: Chi Squared
Distance: 0.005422
```

```
Index: 64
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.004315
```

```
Index: 65
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.004509
```

```
Index: 66
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: pink
-- Matched --
INSTANCE: pink
CATEGORY: pink
ROTATION: 115
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.005719
```

```
Index: 67
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.003144
```

```
Index: 68
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.003144
```

```
Index: 69
-- Original --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: sparkel
-- Matched --
INSTANCE: sparkel
CATEGORY: sparkel
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: sparkel
Distance Metric: Chi Squared
Distance: 0.003517
```

```
Index: 70
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 0
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
Distance Metric: Chi Squared
Distance: 0.003846
```

```
Index: 71
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.006433
```

```
Index: 72
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 45
LIGHTING: 0
MATCHED_CATEGORY: blue
-- Matched --
INSTANCE: blue
CATEGORY: blue
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.002606
```

```
Index: 73
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 180
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.006433
```

```
Index: 74
-- Original --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 90
LIGHTING: 0
MATCHED_CATEGORY: yellow
-- Matched --
INSTANCE: yellow
CATEGORY: yellow
ROTATION: 62
LIGHTING: 0
MATCHED_CATEGORY: yellow
Distance Metric: Chi Squared
Distance: 0.012321
```